

Notes: Morck, Shleifer, and Vishny (JFE, 1988)

Management Ownership and Market Valuation: An Empirical Analysis

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1 Big picture

- **Question.** How is management ownership related to market valuation? More specifically, does Tobin's Q rise monotonically with managerial ownership because managers' incentives become better aligned with shareholders, or can high ownership also make managers entrenched and reduce value?
- **Core tension.** Jensen and Meckling's convergence-of-interests view predicts that higher managerial ownership should increase firm value. Demsetz, Fama, and Jensen emphasize an offsetting entrenchment channel: once managers have enough control, they may be harder to discipline through the labor market, product market, board, or takeover market.
- **Main empirical object.** The paper studies a 1980 cross-section of 371 large Fortune 500 firms and relates Tobin's Q to the fraction of common stock owned by the board of directors.
- **Main finding.** The relationship is strongly nonmonotonic. Q first rises as board ownership moves from negligible levels to about 5%, then falls as board ownership rises from about 5% to 25%, and then rises slightly for still higher ownership stakes.
- **Interpretation.** The authors interpret the low-ownership positive slope as broadly consistent with incentive alignment, the middle-range negative slope as consistent with entrenchment, and the high-ownership positive slope as evidence that once control is already entrenched, additional cash-flow ownership may again improve incentives.
- **Additional finding.** For older firms, Tobin's Q is lower when one of the top two officers is a member of the founding family. For younger firms, founder-family leadership is associated with higher Q , consistent with founders being more valuable in younger entrepreneurial firms and more likely to be entrenched in older ones.
- **Economic takeaway.** Managerial ownership is not simply "good" or "bad." Small stakes may align incentives, intermediate stakes may help managers entrench themselves, and very large stakes may again partly align cash-flow interests.

2 Incremental contribution of the paper

- **Introduces a nonmonotonic ownership-value framework.** Earlier work often searched for a simple linear relation between ownership and performance. This paper shows that linear specifications can miss the central pattern: ownership may have different effects at different ranges.
- **Combines two governance channels.** The paper brings together the convergence-of-interests hypothesis and the entrenchment hypothesis in a single empirical design.
- **Uses market valuation rather than only accounting returns.** Tobin's Q is used as the main measure because it reflects the market's valuation of tangible assets plus intangible assets and future opportunities, not only current accounting profitability.
- **Focuses on board ownership.** The main ownership variable is the common stock owned by the board of directors, rather than total ownership concentration by all large shareholders. This is important because the governance mechanism being studied is management control, not outside blockholder monitoring in general.

- **Separates officers from outside directors.** The paper decomposes board ownership into stakes held by top officers and stakes held by other board members. Both display similar non-monotonic patterns, suggesting that outside directors with large stakes can also matter for governance.
- **Adds founder-family governance.** The paper studies whether control can arise through status and history, not just voting power. Founder-family leadership in old firms is treated as evidence of a possible non-ownership source of entrenchment.
- **Anticipates later corporate governance research.** The paper foreshadows later work on managerial entrenchment, takeover resistance, founding families, board incentives, and nonlinear governance-performance relations.

3 Conceptual framework

3.1 Convergence of interests

The convergence-of-interests hypothesis begins from the Berle and Means separation of ownership and control. When managers own little equity, they bear only a small share of the cost of non-value-maximizing behavior. They may consume perks, shirk, overinvest, build empires, protect employees, or pursue sales growth rather than shareholder value.

As management ownership rises, managers internalize more of the value consequences of their decisions. The pure convergence view therefore predicts:

$$\frac{\partial Q}{\partial \text{managerial ownership}} > 0. \quad (1)$$

Empirical implication. If only convergence of interests matters, Tobin’s Q should rise monotonically with management ownership.

3.2 Entrenchment

The entrenchment view emphasizes that ownership also gives managers control. A manager with enough voting power or influence can become difficult to remove, difficult to monitor, and difficult to discipline through takeovers or the board. Once entrenched, managers can pursue private benefits and non-value-maximizing projects.

This implies that ownership can have a negative effect over some range:

$$\frac{\partial Q}{\partial \text{managerial ownership}} < 0 \quad (2)$$

when the marginal effect of extra ownership is mainly more control rather than better incentives.

Important nuance. Entrenchment need not require 50% voting ownership. A manager can be entrenched through tenure, founder status, influence over directors, personality, family control, or weak outside monitoring.

3.3 Combined prediction

The paper’s key empirical idea is that both channels can operate at once:

Ownership range	Dominant force	Predicted relation with Q
Low ownership	Incentive alignment	Positive
Intermediate ownership	Entrenchment rises	Negative
Very high ownership	Entrenchment may be saturated; cash-flow incentives rise	Positive or weakly positive

3.4 Founder-family control

The authors also ask whether managers can be entrenched for reasons other than equity ownership. A founding family may have status, influence, historical control of the board, and firm-specific authority. But founder-family leadership can have two opposite interpretations:

- **Entrepreneurial-value interpretation.** In young firms, founders or their families may embody key entrepreneurial skill, vision, or firm-specific knowledge.
- **Entrenchment interpretation.** In older firms, founder-family managers may persist because of control and status rather than superior ability.

Thus, founder-family leadership can have different valuation effects depending on firm age.

4 Data and measurement

4.1 Sample construction

- **Universe.** The starting point is a December 1980 Corporate Data Exchange listing of names and stakes of large shareholders for 456 Fortune 500 firms.
- **Board ownership.** CDE identifies shareholders who are members of the board of directors, except for board members whose stakes are below 0.2%.
- **Final sample.** The final sample contains 371 firms for which Tobin's Q can be constructed.
- **Why 371 rather than 456?** Values of Q are unavailable for 85 firms, mainly because of difficulties in obtaining market values of long-term debt or replacement cost of assets.
- **Selection check.** The omitted firms do not appear unusual in observable ownership dimensions: their mean board stake and founder-family presence are close to those in the final sample.

4.2 Dependent variables

- **Tobin's Q .** The main dependent variable is the ratio of market value of the firm to replacement cost of physical assets.
- **Market value.** The numerator is the market value of common stock plus estimated market values of preferred stock and debt.
- **Replacement cost.** The denominator is the replacement cost of plant and inventories, taken from the Griliches R&D master file.
- **Profit rate.** As a comparison with Demsetz and Lehn, the authors also use the profit rate, defined as net cash flows less inflation-adjusted depreciation divided by replacement cost of tangible assets.

4.3 Ownership variables

The primary ownership variable is **BOARD**, the total common-stock stake held by all members of the board of directors. The paper uses piecewise-linear functions of board ownership:

$$BRD.Oto5 = \min(BOARD, 0.05), \quad (3)$$

$$BRD.5to25 = \min\{\max(BOARD - 0.05, 0), 0.20\}, \quad (4)$$

$$BRD.OVER25 = \max(BOARD - 0.25, 0). \quad (5)$$

These variables allow the slope of the ownership- Q relationship to differ across the 0–5%, 5–25%, and above-25% ranges.

4.4 Control variables

The paper includes controls designed to account for observable and unobservable determinants of Tobin’s Q that might also correlate with ownership.

- **R&D per dollar of assets.** Captures intangible capital and growth opportunities.
- **Advertising per dollar of assets.** Captures brand capital and other intangible assets.
- **Long-term debt per dollar of assets.** Captures leverage, tax shields, or profitability-related capital structure.
- **Replacement cost of assets.** Controls for firm size.
- **Three-digit SIC dummies.** Controls for industry-level differences in valuation and ownership structure.

4.5 Officer, outside-board, and founder variables

- **OFFICER.** Combined ownership of the two senior corporate officers, usually the chairman and president.
- **OUTBOARD.** Ownership of the rest of the board. This category can include junior officers, but these individuals usually own little stock.
- **FOUNDER.** Indicator equal to one if a member of the founding family is among the top two officers.
- **INC50.** Indicator equal to one if the firm was incorporated in 1950 or later.
- **FOUNDER50.** Indicator equal to one when both FOUNDER and INC50 are one.

5 Empirical specifications

5.1 Main piecewise-linear model

The benchmark regression can be written schematically as

$$Q_i = \alpha_{industry(i)} + \beta_1 BRD.Oto5_i + \beta_2 BRD.5to25_i + \beta_3 BRD.OVER25_i + X_i' \gamma + \varepsilon_i, \quad (6)$$

where X_i contains R&D/assets, advertising/assets, debt/assets, and replacement cost of assets. Standard errors are heteroskedasticity-consistent.

5.2 Interpretation of slopes

Because board ownership is measured as a fraction, the coefficient is the effect of a 1.00 change in ownership. A 1 percentage point increase corresponds to 0.01 times the coefficient. Thus a coefficient of 6.17 means that increasing board ownership by one percentage point raises Q by about 0.0617 in that ownership region.

5.3 Robustness exercises

The paper performs several checks:

- alternative piecewise breakpoints, such as 2.5%, 7.5%, 15%, 20%, 30%, and a finer grid;
- using 1979 and 1981 Tobin's Q as dependent variables with 1980 ownership variables;
- excluding firms with negligible board ownership;
- excluding oil firms;
- adding growth in the firm's labor force as a control for growth opportunities;
- using profit rate rather than Q ;
- decomposing ownership into officer and outside-board stakes;
- adding founder-family and firm-age variables.

6 Main results

6.1 Board ownership is skewed

The average combined board stake is 10.6%, but the median is only 3.4%. In 28% of sample firms, total board holdings are no more than 1% of shares. In 12% of firms, no board member owns more than 0.2%. At the other end, 31% of firms have board ownership above 10%, 20% have board ownership above 20%, and only 14 firms have board ownership above 50%.

6.2 Raw means suggest nonlinearity

Table 1 groups firms by board ownership. Mean Tobin's Q rises from 0.710 for negligible ownership to 0.879 for the 0–5% group and 0.997 for the 5–10% group. It then declines across parts of the intermediate range, reaching 0.588 in the 25–30% group. Some high-ownership cells are noisy because they contain few observations and a few outliers.

6.3 Piecewise regressions: the central result

With industry controls and firm characteristics included, the main Q regression in Table 2 produces the following slopes:

Ownership range	Estimated slope	Interpretation per 1 percentage point
0–5%	6.17	Q rises by about 0.062
5–25%	-1.60	Q falls by about 0.016
Above 25%	0.794	Q rises by about 0.008

Economic meaning. Moving from negligible ownership to 5% board ownership is associated with a very large increase in Q . Moving from 5% to 25% roughly reverses the increase. Beyond 25%, Q rises again, but much more slowly.

Statistical meaning. The three board-ownership variables are jointly important. The paper reports an F -statistic of 3.77 for the joint hypothesis that the three board-ownership coefficients are zero.

6.4 Interpretation of the nonmonotonic pattern

- **0–5% region.** The positive slope is consistent with incentive alignment: even small ownership stakes may discipline managers.
- **5–25% region.** The negative slope is consistent with entrenchment: ownership in this range may be associated with voting influence, founder status, board control, tenure, or other mechanisms that insulate managers.
- **Above 25%.** The positive slope may indicate that entrenchment is already mostly achieved at roughly 25%, so additional ownership mainly increases managers' cash-flow incentives.

6.5 Profit-rate regressions

The profit-rate regressions show a similar qualitative pattern at low ownership levels, but the estimates are weaker. The 0–5% slope is positive and significant, while the intermediate and high-ownership slopes are not strongly significant. The authors emphasize that a simple linear regression of profit rate on board ownership yields no meaningful relation, consistent with Demsetz and Lehn, because the linear specification averages over offsetting effects.

6.6 Alternative breakpoints and time stability

Table 3 shows that the main pattern is robust to many alternative breakpoints. Specifications with a positive low-ownership slope and negative intermediate slope repeatedly fit the data. The 5% and 25% breakpoints produce the lowest sum of squared errors among the two-breakpoint specifications the authors tried. Using 1979 or 1981 Q as the dependent variable gives similar slopes, suggesting the result is not purely an artifact of one year.

6.7 Wealth constraints and firm size

A potential alternative explanation is that managers can afford a larger percentage stake only in firms with low market value, mechanically producing a negative relation between ownership and Q . Table 4 examines size measures by ownership group. The authors argue that among firms with board ownership of at least 5%, the correlation between board ownership and replacement cost of assets is only -0.02. This weak correlation makes it less likely that wealth constraints alone explain the intermediate-range decline in Q .

The authors also show that excluding the 46 negligible-ownership firms preserves the main pattern. Excluding oil firms also preserves and even strengthens the coefficient pattern.

6.8 Growth opportunities

High- Q firms may also be firms with valuable growth opportunities. If managers in high-growth firms receive more stock, the positive low-ownership slope could be partly spurious. The authors

add the 1970–1980 growth rate of the firm’s labor force. The growth control enters positively, and the ownership coefficients are somewhat smaller, but the nonmonotonic pattern remains.

6.9 Officers and outside board members

Table 5 decomposes ownership into stakes held by the top two officers and stakes held by other board members. Both ownership variables display a similar pattern:

- Q rises as officer ownership moves from 0 to 5%, falls from 5 to 25%, and rises again after 25%.
- Q rises as outside-board ownership moves from 0 to 5%, falls from 5 to 25%, and rises again after 25%.

The estimated effects for outside-board ownership are important because they suggest that outside directors are not merely irrelevant. Their equity stakes may strengthen monitoring incentives, but large stakes may also be associated with their own entrenchment or unchecked influence.

6.10 Founder-family effects

The founder-family regression adds FOUNDER, FOUNDER50, and INC50 to the benchmark model. The main estimates imply:

- Among older firms, founding-family involvement reduces Q by about 0.147, with a t -statistic around -1.91.
- Among newer firms incorporated in 1950 or later, founding-family involvement raises Q by about $-0.147 + 0.614 = 0.467$, with a t -statistic around 2.02.

Interpretation. Founders may be valuable in young firms because entrepreneurial talent matters. In older firms, founder-family control may instead proxy for entrenchment and resistance to value-maximizing discipline.

7 Figures and tables

7.1 Table 1 and Figure 1: Raw means and implied nonlinear relation

Table 1 read:

- The raw group means show that Q is not monotonically related to board ownership.
- The highest raw mean Q among common ownership ranges is around the 5–10% and 15–20% categories, but the cells become sparse and noisy at high ownership levels.
- The table also reveals that high ownership is uncommon in a Fortune 500 sample; many high-stake cells contain fewer than 10 firms.

Figure 1 read:

- The figure plots the fitted piecewise-linear relation from the main Table 2 regression.
- The fitted Q rises sharply from 0 to 5% ownership, falls sharply to about 25%, and then rises more gradually.
- The visual is the paper’s central result: managerial ownership has both incentive and entrenchment associations.

Table 1
Mean values of Tobin's Q for 371 Fortune 500 firms in 1980, grouped by level of equity ownership of the board of directors.

Board's stake	Number of firms	Mean Tobin's Q	Standard error of mean Q
Negligible ^a	46	0.710	0.0566
0–5%	171	0.879	0.0601
5–10%	38	0.997	0.0948
10–15%	27	0.888	0.129
15–20%	15	0.990	0.205
20–25%	22	0.756	0.0804
25–30%	9	0.588	0.0895
30–35%	6	0.658	0.0990
35–40%	11	1.06	0.243
40–45%	6	0.778	0.243
45–50%	6	0.538	0.0991
50–55%	4	0.440	0.0837
55–60%	3	0.428	0.162
60–65%	3	1.46	0.568
65–70%	1	0.283	—
70–75%	2	0.489	0.198
75–80%	1	0.937	—

^aNegligible board stake means that no single member of the board of directors owned more than 0.2% of the firm's common stock.

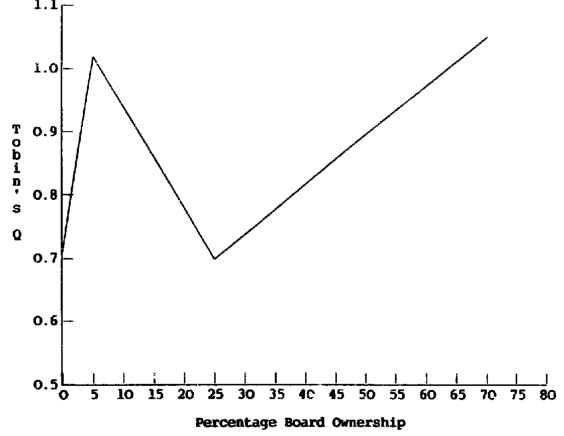


Table 1: Mean Tobin's Q by board ownership group.

Original table extracted from the paper.

Figure 1: Implied relation between board ownership and Tobin's Q .

Original figure extracted from the paper.

7.2 Table 2 and Table 3: Main piecewise regressions and robustness to breakpoints

Table 2 read:

- The benchmark Tobin's Q regression with industry controls gives the main nonmonotonic pattern: positive slope in 0–5%, negative slope in 5–25%, and positive slope above 25%.
- The R&D/assets coefficient is large and positive, consistent with intangible assets being capitalized into Q .
- Long-term debt/assets is negative, consistent with debt being associated with lower Q in this cross-section or with leverage proxying for low profitability.
- Profit-rate regressions preserve the low-ownership positive relation but are weaker overall.

Table 3 read:

- Alternative breakpoints largely confirm that the relationship is nonlinear.
- The strongest pattern is still a sharp rise at low ownership and a decline over a broad intermediate ownership range.
- The 1979 and 1981 results show similar coefficient signs, suggesting the 1980 result is not uniquely tied to one measurement year.

Table 2

Piecewise linear ordinary least squares regressions of 1980 Tobin's Q and profit rate on board ownership and other firm characteristics for 371 Fortune 500 firms.^a

	Dependent variable			
	Tobin's Q	Tobin's Q	Profit rate ^b	Profit rate
Research and development per dollar of assets	—	8.40 ^f (2.23)	—	0.263 ^e (0.115)
Advertising expenses per dollar of assets	—	0.176 (0.816)	—	-0.0264 (0.0759)
Long-term debt per dollar of assets	—	-0.800 ^e (0.365)	—	-0.0664 ^e (0.0298)
Replacement cost (dollar value of assets)	—	-0.0000022 (0.0000038)	—	0.0000004 (0.0000033)
$BRD.0to5^c$	5.74 ^e (2.41)	6.17 ^f (2.04)	0.298 ^e (0.131)	0.328 ^e (0.153)
$BRD.5to25^c$	-1.40 ^d (0.727)	-1.60 ^e (0.638)	-0.0582 (0.0398)	-0.0558 (0.0479)
$BRD.OVER25^c$	-0.0494 (0.458)	0.794 ^d (0.405)	-0.0100 (0.0442)	-0.00153 (0.0442)
Industry dummies for three-digit SIC codes	No	Yes	No	Yes
Number of firms in regression	371	371	315	315
R^2	0.0196	0.595	0.0186	0.429

^aNumbers in parentheses are consistent standard errors calculated according to White (1980).

^bThe profit rate is defined as the ratio of the firm's net cash flows (less the inflation adjusted value of depreciation) to the replacement cost of the firm's tangible assets.

^c $BRD.0to5$ = board ownership if board ownership < 0.05, = 0.05 if board ownership ≥ 0.05. $BRD.5to25$ = 0 if board ownership < 0.05, = board ownership minus 0.05 if 0.05 ≤ board ownership < 0.25, = 0.20 if board ownership ≥ 0.25. $BRD.OVER25$ = 0 if board ownership < 0.25, = board ownership minus 0.25 if board ownership ≥ 0.25.

^dSignificant at 90% confidence level.

^eSignificant at 95% confidence level.

^fSignificant at 99% confidence level.

Table 3

Alternative piecewise linear specifications of 1979, 1980, and 1981 Tobin's Q on board ownership and other firm characteristics for 371 Fortune 500 firms in 1980.^a

Panel A: The dependent variable is 1980 Tobin's Q	
(1) Q = Control variables + 8.75 · $BRD.0to2.5$ - 0.800 · $BRD.2.5to25$ + 0.594 · $BRD.OVER25$ (3.63) (0.533) (0.396)	$R^2 = 0.588, N = 371$
(2) Q = Control variables + 3.78 · $BRD.0to7.5$ - 1.93 · $BRD.7.5to25$ + 0.852 · $BRD.OVER25$ (1.33) (0.672) (0.402)	$R^2 = 0.593, N = 371$
(3) Q = Control variables + 6.67 · $BRD.0to5$ - 2.82 · $BRD.5to15$ + 0.371 · $BRD.OVER15$ (2.48) (1.63) (0.353)	$R^2 = 0.593, N = 371$
(4) Q = Control variables + 6.27 · $BRD.0to5$ - 1.94 · $BRD.5to20$ + 0.541 · $BRD.OVER20$ (2.18) (0.919) (0.372)	$R^2 = 0.593, N = 371$
(5) Q = Control variables + 5.81 · $BRD.0to5$ - 1.27 · $BRD.5to30$ + 0.992 · $BRD.OVER30$ (1.96) (0.529) (0.482)	$R^2 = 0.594, N = 371$
(6) Q = Control variables + 6.99 · $BRD.0to5$ - 3.80 · $BRD.5to10$ - 0.868 · $BRD.10to15$ (2.92) (3.58) (5.36) + 4.83 · $BRD.15to20$ - 10.71 · $BRD.20to25$ + 5.53 · $BRD.25to30$ (6.24) (3.78) (4.78) - 0.220 · $BRD.30to35$ + 0.672 · $BRD.OVER35$ (4.73) (0.623)	$R^2 = 0.599, N = 371$
Panel B: The dependent variable is 1979 Tobin's Q	
Q = Control variables + 5.56 · $BRD.0to5$ - 1.37 · $BRD.5to25$ + 0.702 · $BRD.OVER25$ (1.64) (0.532) (0.302)	$R^2 = 0.637, N = 371$
Panel C: The dependent variable is 1981 Tobin's Q	
Q = Control variables + 4.43 · $BRD.0to5$ - 1.21 · $BRD.5to25$ + 0.665 · $BRD.OVER25$ (1.54) (0.547) (0.455)	$R^2 = 0.567, N = 370$

^aNumbers in parentheses are consistent standard errors calculated according to White (1980).

Table 2: Piecewise regressions of Q and profit rate on board ownership.

Original table extracted from the paper.

Table 3: Alternative breakpoints and adjacent-year Q checks.

Original table extracted from the paper.

7.3 Table 4 and Table 5: Size checks and board-composition regressions

Table 4 read:

- The table examines whether high ownership simply reflects smaller firms in which managers can afford a larger percentage stake.
- Very large firms are concentrated in the negligible ownership group, but among firms with meaningful ownership, size does not decline strongly with board ownership.
- This weakens the claim that the negative intermediate ownership- Q relation is purely a wealth-constraint artifact.

Table 5 read:

- Officer ownership and outside-board ownership each display the same broad nonmonotonic pattern as total board ownership.
- The similarity suggests that both top officers and non-officer directors can have incentive effects at low stakes and entrenchment-related associations at intermediate stakes.
- The results push against a view in which outside directors are entirely irrelevant or merely passive, though they do not prove a causal monitoring channel.

Table 4
Measures of firm size for a 1980 sample of 371 Fortune 500 firms, grouped according to fractional equity ownership of the board of directors.^a

Board's equity stake	Number of firms	Mean replacement cost of assets (in millions)	Mean market value of the firm (in millions)	Mean market value of equity outstanding (in millions)
Negligible ^b	46	\$9134 (1798)	\$6795 (1417)	\$5654 (1284)
0–5%	171	2194 (246)	1705 (195)	1407 (176)
5–10%	38	992 (163)	918 (143)	779 (142)
10–15%	27	2088 (884)	1374 (456)	1141 (369)
15–20%	15	1215 (393)	1043 (341)	878 (277)
20–25%	22	1693 (422)	1287 (354)	1005 (247)
25–30%	9	564 (116)	345 (83)	290 (74)
30–35%	6	3323 (1825)	2177 (1046)	1842 (958)
35–40%	11	1697 (1029)	1741 (886)	1409 (647)
40–45%	6	4815 (3737)	2297 (1334)	1136 (402)
45–50%	6	798 (220)	506 (185)	469 (183)
> 50%	14	458 (118)	296 (96)	257 (87)

^aNumbers in parentheses are standard errors of the means.

^bThe board's equity stake is negligible when no board member is included by Corporate Data Exchange (CDE) in its list of shareholders. That list typically excludes shareholders with less than 0.2% of the firm.

Table 4: Firm size measures by board ownership group.

Original table extracted from the paper.

Table 5
Piecewise linear ordinary least squares regressions of 1980 Tobin's Q on officer and outside board ownership and other firm characteristics.^a

	Dependent variable	
	Tobin's Q	Tobin's Q
Research and development per dollar of assets	—	8.25 ^f (2.27)
Advertising expenses per dollar of assets	—	0.0858 (0.834)
Long-term debt per dollar of assets	—	−0.912 ^e (0.390)
Replacement cost (dollar value of assets)	—	−0.0000031 (0.0000039)
<i>OFF.0to5</i> ^b	3.37 (2.85)	3.98 ^d (2.29)
<i>OFF.5to25</i> ^b	−0.896 (0.933)	−1.56 ^e (0.758)
<i>OFF.OVER25</i> ^b	−0.442 (0.354)	0.775 ^e (0.324)
<i>OUT.0to5</i> ^c	3.26 (2.19)	4.34 ^d (2.29)
<i>OUT.5to25</i> ^c	−2.03 ^e (0.858)	−1.42 ^d (0.779)
<i>OUT.OVER25</i> ^c	1.84 (1.38)	1.69 (1.04)
Industry dummies for three-digit SIC codes	No	Yes
Number of observations	371	371
R^2	0.0215	0.597

^aNumbers in parentheses are consistent standard errors calculated according to White (1980).

^b*OFF.0to5* = combined ownership of top two officers if combined ownership < 0.05, = 0.05 if combined ownership ≥ 0.05. *OFF.5to25* = 0 if combined ownership < 0.05, = combined ownership minus 0.05 if 0.05 ≤ combined ownership < 0.25, = 0.20 if combined ownership ≥ 0.25. *OFF.OVER25* = 0 if combined ownership < 0.25, = combined ownership minus 0.25 if combined ownership ≥ 0.25.

^cDefined for combined ownership of the directors who are not top officers analogously to the officer variables.

^dSignificant at 90% confidence level.

^eSignificant at 95% confidence level.

^fSignificant at 99% confidence level.

Table 5: Officer and outside-board ownership regressions.

Original table extracted from the paper.

8 Economic interpretation

8.1 What the paper says about ownership incentives

The sharp positive association between Q and board ownership at very low ownership levels is consistent with the idea that small managerial stakes can have large incentive effects. However, the authors are careful not to interpret this purely as causal. High- Q firms may give managers more stock, managers of high- Q firms may retain more shares, or high ownership may signal managerial confidence.

8.2 What the paper says about entrenchment

The negative association between Q and ownership in the 5–25% range is the most direct evidence for the entrenchment idea. The authors do not claim that 5% ownership gives enough voting power to guarantee control. Instead, ownership may be correlated with other entrenching attributes: founder status, board influence, tenure, and dominance of insiders over outsiders.

8.3 Why Q rises again after 25%

The positive slope beyond 25% suggests that once effective control is already strong, additional ownership may not add much further entrenchment. At that point, the additional cash-flow stake

may again align managers with shareholders. Another way to say this is that the marginal control benefit of ownership may be low after 25%, while the marginal incentive effect remains positive.

8.4 How to interpret the founder-family result

The founder-family result is important because it separates ownership from control status. In old firms, founding-family leadership is associated with lower Q even after controlling for ownership. This supports the idea that managerial entrenchment can come from social, historical, and organizational sources, not just stock ownership.

9 Limitations and cautions

- **Cross-sectional identification.** The paper documents robust associations, not clean causal effects. Ownership structure can be chosen in response to firm characteristics.
- **Tobin's Q is noisy.** Q captures market value relative to replacement cost, but it also reflects intangible assets, growth opportunities, industry conditions, and measurement error.
- **Large-firm sample.** The sample is Fortune 500 firms. Results may differ for small, young, high-growth firms in which founders and stock ownership play different roles.
- **Ownership threshold.** CDE omits individual board stakes below 0.2%, so very small stakes are not fully measured.
- **Board heterogeneity is simplified.** The paper uses aggregate board, officer, and outside-board stakes. It cannot fully model bargaining, coalitions, or influence among individual directors.
- **Compensation is incomplete.** Officer stock ownership is only part of managerial incentives. Salary, bonus, options, pensions, and career concerns also matter.
- **Private benefits and efficiency.** A lower Q with entrenchment does not automatically prove social inefficiency. If managers derive private benefits from control, the welfare analysis depends on whether side payments and control allocations are efficient.

10 Conclusion

- **Core conclusion.** Management ownership and firm value have a nonmonotonic relation. Tobin's Q rises at low board ownership, falls at intermediate ownership, and rises slightly again at high ownership.
- **Economic intuition.** The pattern is consistent with a tradeoff between incentive alignment and entrenchment. Low ownership improves incentives; intermediate ownership may facilitate control; very high ownership may mainly increase cash-flow alignment once entrenchment is already saturated.
- **Incremental contribution revisited.** The paper's main contribution is not only that ownership matters, but that the sign of the ownership-value relation depends on the ownership range and on who controls the firm.

- **Founder-family implication.** Founder-family leadership appears beneficial in younger firms but detrimental in older firms, consistent with a shift from entrepreneurial value to possible entrenchment as firms age.
- **Governance implication.** Corporate governance cannot be summarized by a single linear measure of ownership concentration. The identity of owners, the ownership range, the control environment, and historical sources of authority all matter.
- **Takeaway.** The paper provides one of the classic empirical foundations for the view that managerial ownership has both an incentive-alignment side and an entrenchment side.